

Test 1

Yadibere Clemie

05 mai 2026

IMG 6203.jpeg

Caveman 3

$$\begin{aligned} \int \frac{dx}{x\sqrt{1+x^8}} &= \int \frac{dx}{x\sqrt{x^8(\frac{1}{x^8}+1)}} = \int \frac{dx}{x \cdot x^4 \sqrt{x^{-8}+1}} \\ &= \int \frac{dx}{x^5 \sqrt{x^{-8}+1}} = \int \frac{x^{-5} dx}{\sqrt{x^{-8}+1}} \\ u = x^{-4} &\rightarrow u^2 = x^{-8} \end{aligned}$$

$$du = -4x^{-5} dx$$

$$-\frac{1}{4} du = x^{-5} dx$$

$$u = \tan \theta$$

$$du = \sec^2 \theta d\theta$$

I

$$\begin{aligned} -\frac{1}{4} \int \frac{\sec^2 \theta}{\sqrt{\tan^2 \theta + 1}} d\theta &= -\frac{1}{4} \int \frac{\sec^2 \theta}{\sqrt{\sec^2 \theta}} d\theta \\ &= -\frac{1}{4} \int \frac{\sec^2 \theta}{\sec \theta} d\theta = -\frac{1}{4} \int \sec \theta d\theta \\ &= -\frac{1}{4} \ln |\sec \theta + \tan \theta| + C \\ &= -\frac{1}{4} \ln |\sqrt{x^{-8}+1} + x^{-4}| + C \end{aligned}$$

$$\sec \theta = \sqrt{u^2 + 1}$$

$$= \sqrt{x^{-8} + 1}$$

$$\tan \theta = u$$

$$= x^{-4}$$

IMG 6204.jpeg

$$\begin{aligned} \int \frac{dx}{x^{\frac{169}{169}} x^{\frac{144}{169}} + x^{\frac{25}{169}}} &= \int \frac{dx}{x^{\frac{25}{169}} \left(x^{\frac{144}{169}} \cdot x^{\frac{44}{169}} + 1 \right)} \\ &= \int \frac{dx}{x^{\frac{25}{169}} \left(\left(x^{\frac{144}{169}} \right)^2 + 1 \right)} \Rightarrow du = x^{144/169} \\ &= \int \frac{x^{\frac{169}{169}} dx}{\frac{169}{144} du = x^{-25/169} dx} \\ &\left(\left(x^{\frac{144}{169}} \right)^2 + 1 \right) \\ \frac{169}{144} \int \frac{du}{u^2+1} &= \frac{169}{144} \arctan(u) + C \\ \frac{169}{144} \arctan \left(x^{144/169} \right) &+ C \end{aligned}$$

IMG 6202.jpeg

Caveman Method 2

$$\begin{aligned} & \int \frac{x^2+1}{x^4+1} dx \\ & \int \frac{\frac{x^2+1}{x^2}}{\frac{x^4+1}{x^2}} dx = \int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}} dx \\ & \left(\begin{array}{l} du = 1 + \frac{1}{x^2} dx = 1 + x^{-2} dx \\ u = x - x^{-1} \\ \left(x - \frac{1}{x}\right)^2 = x^2 - 2 + \frac{1}{x^2} \end{array} \right) \\ & \int \frac{1+\frac{1}{x^2}}{x^2-2+\frac{1}{x^2}+2} dx \\ & = \int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2} + 2 dx \\ & = \int \frac{1}{u^2+2} du = \frac{1}{\sqrt{2}} \arctan \frac{u}{\sqrt{2}} + C \\ & = \frac{1}{\sqrt{2}} \arctan \frac{x-\frac{1}{x}}{\sqrt{2}} + C \end{aligned}$$

IMG 6201.jpeg

Caveman Method

$$\begin{aligned} & \int \frac{1}{x^{200}+x} dx \\ & \int \frac{1/x^{200}}{\frac{x^{200}+x}{x^{200}}} dx = \int \frac{x^{-200}}{1+x^{-199}} dx \\ & \quad u = 1 + x^{-199} \\ & \quad du = -199x^{-200} dx = -\frac{1}{199} \int \frac{1}{u} du \\ & -\frac{1}{199} du = x^{-200} dx = -\frac{1}{199} \ln |u| + C \\ & \quad = -\frac{1}{199} \ln |1 + x^{-199}| + C \\ & \quad = -\frac{1}{199} \ln \left| \frac{x^{199} + 1}{x^{199}} \right| + C \end{aligned}$$